

2-4 Use Addition Properties to Add

Learning goal: I can change the order of addends or group them in a different way to add more easily, and I know the sum stays the same.

Textbook: pages 47–50. The book adds $27 + 53 + 40$ by grouping and by changing the order.

Words to Know

addend	A number you add. In $34 + 66$, the addends are 34 and 66.
sum	The answer when you add.
equation	A number sentence with an equal sign. Both sides have the same value.
order property	Switch the order of addends and the sum stays the same: $18 + 36 = 36 + 18$. (Grown-up name: Commutative Property.)
grouping property	Choose which addends to add first and the sum stays the same. (Grown-up name: Associative Property.)
friendly pair	Two addends whose sum ends in 0 or 00, like $46 + 54 = 100$.

How to Teach It

- Show that order does not matter: $4 + 3$ and $3 + 4$ are both 7. Then try bigger numbers: $245 + 132 = 132 + 245$.
 - For a fill-in equation like $245 + 132 = 132 + \underline{\quad}$, look at both sides. The same addends must be there, so the blank is 245. No adding is needed.
 - With three addends, look for a **friendly pair** (ones digits that make 10, or numbers that make a hundred). Draw a V from the two addends to their sum.
 - Add the friendly pair first, then add the third number. If the pair is not next to each other, switch the order first.
 - To check an expression, it must use the **same numbers** and **only addition**.
- Tip:** Ask, “Which two numbers are easy to add together?” Ones digits that make 10 ($7 + 3$, $8 + 2$) are a great clue.

Worked Examples

Example 1: Group a friendly pair

$$34 + 66 + 25$$



$$100 + 25 = 125$$

4 + 6 make a ten, so add $34 + 66 = 100$ first. Sum: **125**.

Example 2: Switch the order, then group

Start: $18 + 45 + 82$. Switch: $18 + 82 + 45$.

$$18 + 82 + 45$$



$$100 + 45 = 145$$

Sum: **145**.

Example 3: Make the equation true

$$245 + 132 = 132 + \underline{\quad}$$

Both sides need the same addends. 132 is already there, so the blank is **245**.

Example 4: Which expressions show the total?

Days: 73, 55, 27.

Yes: $27 + 55 + 73$ and $73 + 27 + 55$ (same numbers, only +).

No: $55 - 27 + 73$ (subtracts) and $73 + 55 + 72$ (72 is not 27).

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|---|---|
| ☐ Explain that the order of addends does not change the sum. | ☐ Group addends to add more easily and find the sum. |
| ☐ Fill in an equation like $324 + 158 = 158 + \underline{\quad}$ without adding. | ☐ Choose all expressions that give the same total. |
| ☐ Find a friendly pair in three addends. | ☐ Explain why a way of grouping is easier. |

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Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Make the equation true. Do you need to add?

$$436 + 251 = 251 + \underline{\hspace{2cm}}$$

- B.** The table shows how many cans a class collected. Circle **all** the expressions that find the total.

A. $36 + 64 + 49$

B. $64 + 49 - 36$

C. $49 + 36 + 64$

D. $64 + 94 + 36$

Day	Cans
Saturday	64
Sunday	49
Monday	36

- C.** How can you group the addends to make it easier? Find the sum.

$$46 + 125 + 54$$

Sum: _____

On My Own *Try these by yourself. Show your work.*

- 1.** Make the equation true.

$$347 + 286 = 286 + \underline{\hspace{2cm}}$$

- 2.** Make the equation true.

$$518 + \underline{\hspace{2cm}} = 69 + 518$$

- 3.** Make the equation true.

$$624 + 153 = \underline{\hspace{2cm}} + 624$$

- 4.** Make the equation true.

$$\underline{\hspace{2cm}} + 87 = 87 + 702$$

- 5.** How can you group the addends to make it easier to find the sum? Explain your thinking.

$$263 + 418 + 137$$

Sum: _____

- 6.** Show one way to group the addends and solve.

$$165 + 28 + 72$$

- 7.** Show one way to group the addends and solve.

$$396 + 204 + 172$$

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- 8.** Show one way to group the addends and solve.

$$243 + 316 + 57$$

- 9.** Show one way to group the addends and solve.

$$125 + 314 + 175$$

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- 10.** Sam buys a bike for \$276, a helmet for \$145, and a lock for \$124. How can he group the prices to find the total cost? What is the total?

Total: _____

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- 11.** Jada says $58 + 39 + 42$ is easier if she adds $58 + 42$ first. Is she right? Explain, and find the sum.

Sum: _____

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Together

A	436 <i>No adding needed: both sides must have the same two addends (order property).</i>
B	Circle A and C . <i>B subtracts. D uses 94 instead of 49. The total is 149.</i>
C	$46 + 54 = 100$, then $100 + 125 = \mathbf{225}$ <i>6 + 4 make a ten, so 46 and 54 are a friendly pair. Also accept adding left to right if the sum is 225.</i>

On My Own

1	347
2	69
3	153
4	702
5	Group $263 + 137 = 400$ first (3 + 7 make a ten). Then $400 + 418 = \mathbf{818}$. <i>Accept any explanation that adds a friendly pair first.</i>
6	$28 + 72 = 100$; $100 + 165 = \mathbf{265}$ <i>Items 6–9: accept any grouping with the correct sum; praise a friendly-pair grouping.</i>
7	$396 + 204 = 600$; $600 + 172 = \mathbf{772}$
8	$243 + 57 = 300$; $300 + 316 = \mathbf{616}$
9	$125 + 175 = 300$; $300 + 314 = \mathbf{614}$
10	Group $\$276 + \$124 = \$400$ first. Then $\$400 + \$145 = \mathbf{\$545}$. <i>Other groupings are acceptable if the total is \$545.</i>
11	Yes. $58 + 42 = 100$ because 8 + 2 make a ten. Then $100 + 39 = \mathbf{139}$. <i>She can add in any order or grouping and the sum stays the same.</i>